Sequence 4

REAL NUMBERS AND INTERVALS

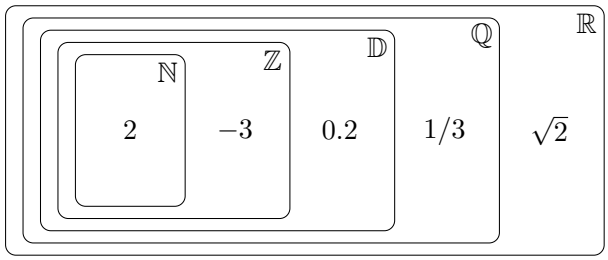
2–13 November 2026

I Sets of numbers

Definitions

- $\mathbb{N}$: natural numbers; $\mathbb{Z}$: integers.
- $\mathbb{D}$: decimal numbers, of the form $a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$.
- $\mathbb{Q}$: rational numbers, of the form a/b with $a, b \in \mathbb{Z}$, $b \neq 0$.
- $\mathbb{R}$: real numbers, represented by the abscissae of the points of a graduated line. A real number that is not rational is **irrational**.

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{D} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}.$$



Example 4.1

Give the smallest of the five sets containing each of the numbers: -95; 7.29; $5/3$; $\sqrt{25}$; $\sqrt{2}$; π; 0.0002.

Solution:

II Two proofs by contradiction

Proof by contradiction

To prove a statement, assume its negation, deduce a contradiction, then conclude that the statement is true.

Property

$\frac{1}{3}$ is not a decimal number.

Proof to write.

Solution:

Property

$\sqrt{2}$ is irrational.

Proof to write.

Solution:

III Intervals of the real line

Notation

For two real numbers $a < b$, intervals describe the real numbers lying between two endpoints, with or without the endpoints. A closed bracket includes the endpoint, an open bracket excludes it. The symbols $-\infty, +\infty$ are not real numbers and are always paired with an open bracket.

Condition	Interval	Description
$a \leq x \leq b$	$[a; b]$	both endpoints included
$a < x \leq b$	$]a; b]$	a excluded, b included
$a \leq x < b$	$[a; b[$	a included, b excluded
$a < x < b$	$]a; b[$	both endpoints excluded
$x \geq a$	$[a; +\infty[$	greater than or equal to a
$x > a$	$]a; +\infty[$	strictly greater than a
$x \leq b$	$]-\infty; b]$	less than or equal to b
$x < b$	$]-\infty; b[$	strictly less than b

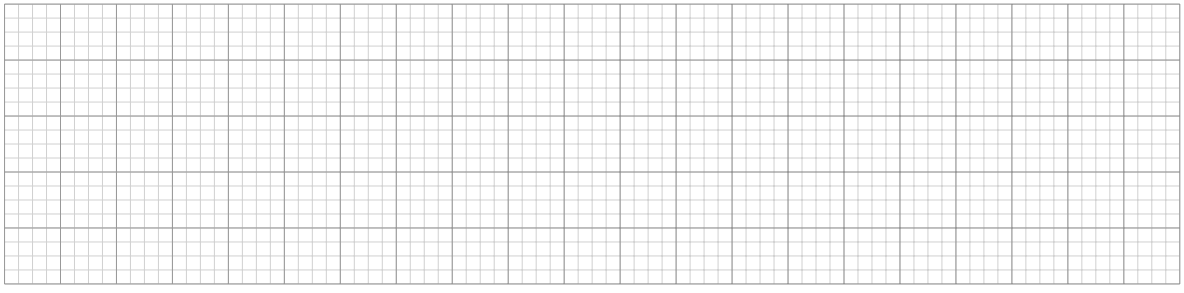
$$\mathbb{R} =]-\infty; +\infty[, \quad \mathbb{R}^* =]-\infty; 0[\cup]0; +\infty[.$$

Watch out. The strictly negative real numbers form $]-\infty; 0[$: 0 is excluded. $[a; a] = \{a\}$; $]a; a[= \varnothing$.

Example 4.2

Write as an interval then represent: $2 \leq x \leq 5$; $x > 4$; $-6 < x \leq 2$; $x < 0$.

Solution:



IV Intersection and union of intervals

Definition

$I \cap J$ corresponds to the condition "belonging to I **and** to J"; $I \cup J$ corresponds to "belonging to I **or** to J". A union of intervals is not always an interval.

Example 4.3

With $I = [0; 12]$ and $J = [3; 20]$, determine $I \cap J$ and $I \cup J$.

Solution:

Example 4.4

Complete, justifying with representations on a number line.

I	J	$I \cap J$	$I \cup J$
$]-4; 3]$	$[1; 5]$	.....	.....
$]-\infty; 2]$	$[-4; +\infty[$	.....	.....
$]-\infty; 7]$	$[7; +\infty[$	.....	.....
$]-\infty; 5]$	$[-3; +\infty[$	.....	.....
$]-2; 0[$	$]0; 3]$	.....	.....

Solution:

Fluency – 5 to 10 minutes

Simplify $\overrightarrow{AB} + \overrightarrow{BC}$; solve $3x - 6 = 0$; calculate 10^{-2}.

V Absolute value and distance

Definition and properties

$|x|$ is the distance from the real number x to 0; $|x| \geq 0$ and

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

The distance between a and b is $|b-a|$. For $a \in \mathbb{R}$ and $r \geq 0$:

$$|x-a| \leq r \iff x \in [a-r; a+r], \qquad |x-a| < r \iff x \in]a-r; a+r[.$$

Example 4.5

Calculate $|3.2|$, $|-1.6|$, $|\sqrt{3}-2|$, then the distances between 2 and 5, 2 and -5, -2 and -5.

Solution:

Example 4.6

Solve: $|x-7/3| = 14$; $|x-2| \leq 3$; $|x+3| < 5$.

Solution:

Example 4.7

Write $x \in [4; 10]$ as an inequality with an absolute value.

Solution:

VI Bounds and approximations

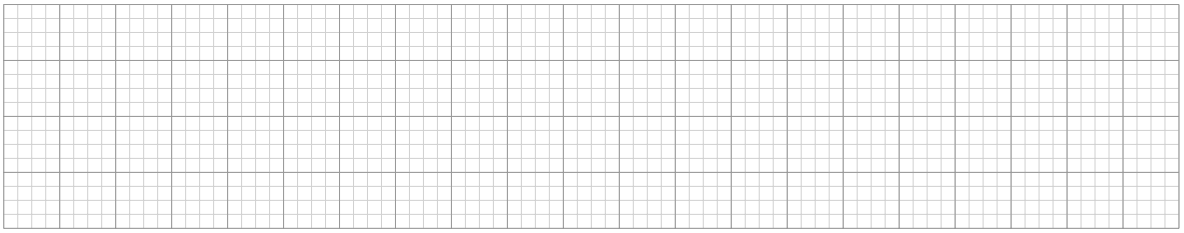
Vocabulary

To bound x is to give $a \leq x \leq b$; the width is $b - a$. An approximation from below is less than or equal to x; an approximation from above is greater than or equal to it. Rounding chooses the nearest value at the required precision. Significant figures are counted from the first nonzero digit; the displayed precision must suit the context.

Example 4.8

Bound $\sqrt{2}$ between two consecutive hundredths. Give its rounding to the nearest hundredth. Round 0.004567 to three significant figures, then 12345 to three significant figures in scientific notation.

Solution:



```
def encadre_racine2(p):
    # p entier naturel ; calculs entiers exacts
    q = 10**p
    k = 0
    while (k+1)**2 <= 2*q*q:
        k = k+1
    return k/q, (k+1)/q
```

Example 4.9

Explain why this function bounds $\sqrt{2}$. What does it return for $p = 2$? Why does it terminate?

Solution:

